WAP to Implement Composite Three-Dimensional Transformations Using Translation, Rotation, and Scaling.

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<math.h>

#define PI 3.14159

void drawcube(float x[], float y[], float z[])
{
   int i;
   int px[8], py[8];

   for(i=0; i<8; i++)
   {
      px[i] = (int)(x[i] + z[i]/2);
      py[i] = (int)(y[i] - z[i]/2);
   }

   for(i=0; i<4; i++)
   {
      line(px[i],py[i],
           px[(i+1)%4],py[(i+1)%4]);

      line(px[i+4],py[i+4],
           px[((i+1)%4)+4],
           py[((i+1)%4)+4]);

      line(px[i],py[i],px[i+4],py[i+4]);
   }
}

void main()
{
   int gd=DETECT,gm;
   int i;
   float x[8]={200,300,300,200,
               200,300,300,200};

   float y[8]={180,180,280,280,
               180,180,280,280};

   float z[8]={0,0,0,0,80,80,80,80};

   float nx[8],ny[8],nz[8];
   float tx,ty,tz;
   float sx,sy,sz;
   float angle,rad;
   float cx=250,cy=230,cz=40;
   float a,b;
```

```
    clrscr();

    printf("Enter translation tx ty tz: ");
    scanf("%f%f%f",&tx,&ty,&tz);

    printf("Enter rotation angle: ");
    scanf("%f",&angle);

    printf("Enter scaling sx sy sz: ");
    scanf("%f%f%f",&sx,&sy,&sz);

    rad=angle*PI/180.0;

    for(i=0;i<8;i++)
    {

       /*Scaling about cube center */

       a=(x[i]-cx)*sx;
       b=(y[i]-cy)*sy;

       /* Rotation about Z-axis */

       nx[i]=cx+a*cos(rad)-b*sin(rad);
       ny[i]=cy+a*sin(rad)+b*cos(rad);

       nz[i]=cz+(z[i]-cz)*sz;

       /* Translation */

       nx[i]=nx[i]+tx;
       ny[i]=ny[i]+ty;
       nz[i]=nz[i]+tz;
    }

    initgraph(&gd,&gm,"C:/TURBO C3/BGI");

    setcolor(WHITE);
    drawcube(x,y,z);

    setcolor(YELLOW);
    drawcube(nx,ny,nz);

    outtextxy(10,20,"WHITE: Original Cube");
    outtextxy(10,40,"YELLOW: Composite Transformation");

    getch();
    closegraph();
}
```